

Grazing and browsing

Vast grasslands cover one-fifth of Earth's land surface. They are home to grazers, including many of the world's largest mammals, which fuel their bodies on a diet of pure grass.

Although grasses grow in abundance, they are low in energy and nutrients, and they are difficult to digest. To survive, grazers spend much of the day feeding. Browsing animals are selective feeders and they gather the leaves, buds, fruit, or seeds of particular plants to eat.

Prairie giant

The American bison is a heavyweight grazer with a big appetite—adult males need to consume up to 110 lb (50 kg) of grass a day. Like many other large grazing mammals, the bison is a ruminant with a special digestive system for breaking down its tough, fibrous food. At one time, giant herds of more than a million bison roamed the North American prairies. Today, however, their numbers have shrunk dramatically, due to overhunting and to people plowing up too much of the grasslands to grow crops.

Long, curved claws clamp limbs to branches, letting sloth climb or hang without tiring

Geese store body fat, converted from grass in their diet

Grazing geese

Mammals are not the only large animals that feed by grazing. These pink-footed geese are eating grass during a migration stopover near England's east coast. Although they swim well and have webbed feet, geese gather most of their food on land. The birds have no teeth, but they can tear up grass by gripping clumps in their beaks and then giving them a sharp tug.



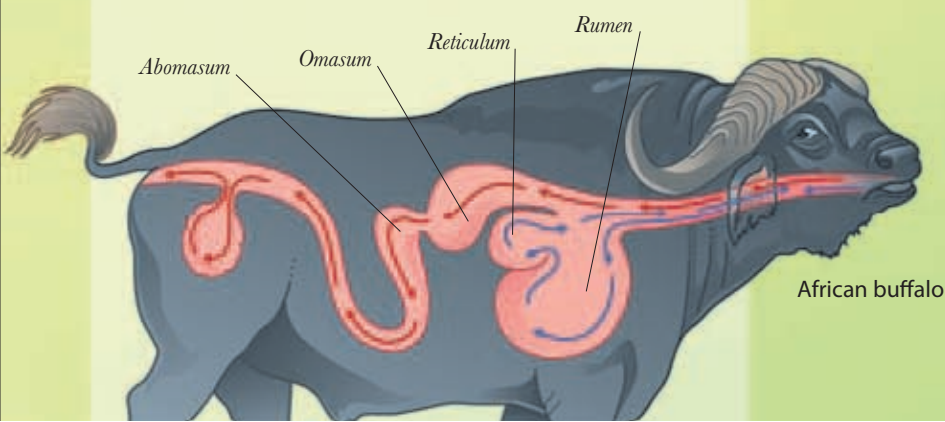
Digesting plant food

Ruminant mammals, such as sheep and cattle, cannot digest the tough substance in grass called cellulose. They get help from microorganisms, which live in the rumen—the largest of their four stomach chambers. Here, the microorganisms break down the cellulose, and the animal then regurgitates its food to chew it for a second time. The food then travels through the other stomach chambers, where its nutrients are absorbed.

Slow-motion meal

Clinging tightly to a branch with its hook-shaped claws, a two-toed sloth leisurely chews its way through a leaf.

Sloths spend almost all of their time in trees and are legendary for their unhurried movements, spending many hours resting each day. A sloth digests its food equally slowly. It can take more than a month for leaves to pass through its body—a record digestion time for any plant-eating mammal.



African buffalo